PROPOSAL DEFENSE · JULY, 2026

NEXIAM

A Role-Based College Admission Testing System for the Guidance and Registrar Offices at ISPSC Tagudin

SARMIENTO, D. D. · PAGADUAN, J. G. · LOPEZ, C. M.

BSIT — ISPSC Tagudin Campus | Adviser: Dr. Zeus Vincent Mendoza

THE PROBLEM — GLOBAL & NATIONAL

A Worldwide Shift, A Familiar Strain

Paper-based testing → slow, integrity risks, equity gaps

1. Global shift to AI-assisted, cloud-based admission platforms
   → ML applicant assessment validated (Lira et al., Science Advances)

2. RA 10931 increased admission volumes, strained staff and infrastructure
   → Unified platforms proven deployable across institutions (Adwan et al., 97% usability)

3. RA 10173 — Data Privacy Act — data protection is a legal duty
   → Regulatory floor governing all system design

THE PROBLEM — AT ISPSC TAGUDIN

Same Pressure, Paper Process

700–1,000

applicants processed every year —
entirely on paper

1. Hand-scoring delay — five days, sometimes up to two full weeks per cycle.
2. Results posted on a Facebook page, monitored during office hours only.
3. Inter-office coordination runs on text messages — no shared system or visibility.
4. Gap: components proven independently, integration missing
   → RAG chatbot: Dayata & Yap (IEEE COMNETSAT)
   → OMR scoring: Cuerdo (PH public schools)
   → ML-based decision-support: Tan (ACM)
   * **No unified admission platform exists** *

OBJECTIVES OF THE STUDY

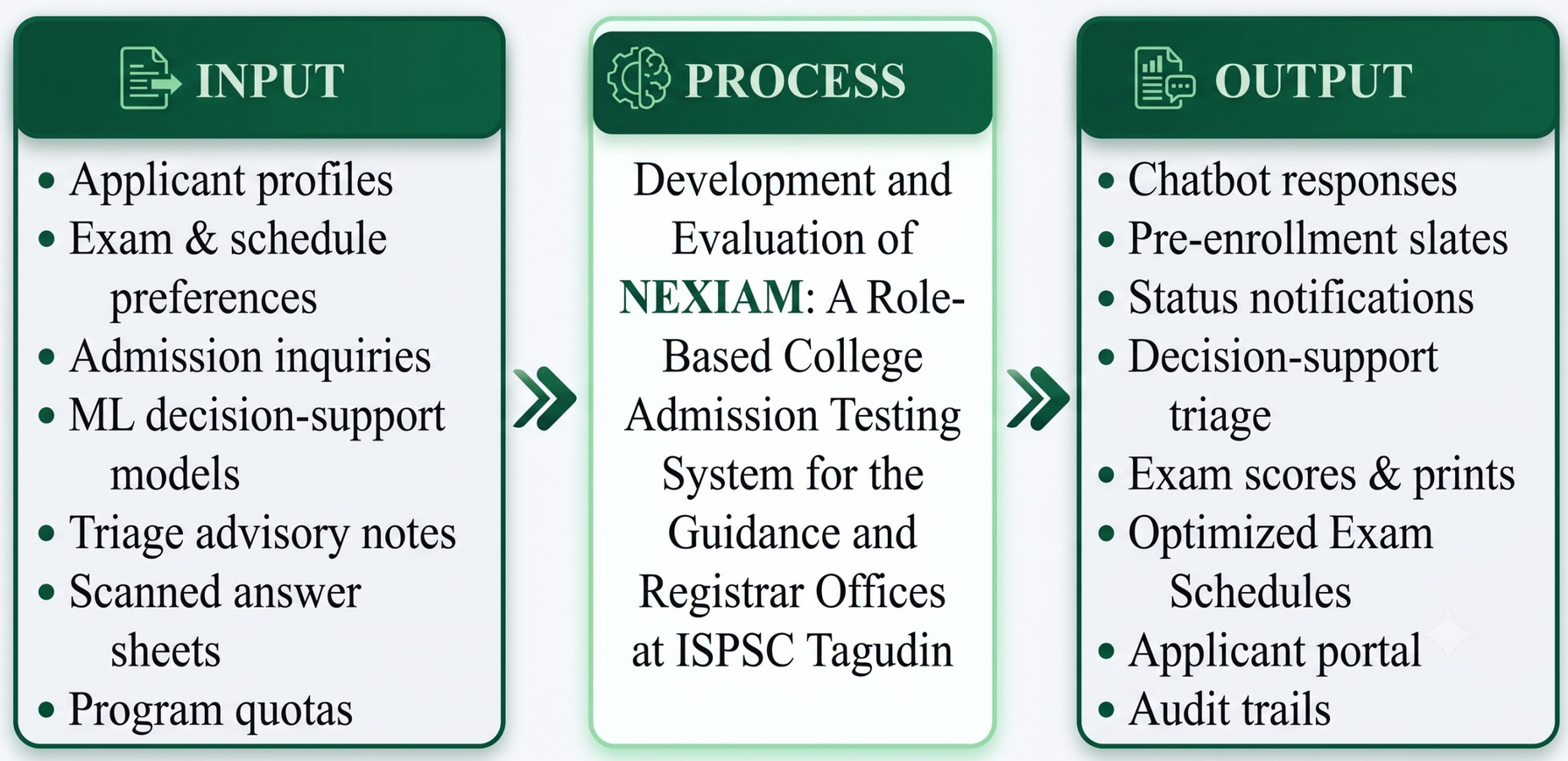
One General Objective, Four Specific Objectives

To Develop NEXIAM: A Role-Based College Admission Testing System for the Guidance and Registrar Offices at ISPSC Tagudin Campus.

- **SO1** Analyze the existing admission process and pinpoint its gaps.
- **SO2** Determine the system requirements those gaps demand.
- **SO3** Evaluate user acceptance of NEXIAM using the Technology Acceptance Model.
- **SO4** Test whether NEXIAM performs measurably better than the manual process it replaces.

CONCEPTUAL FRAMEWORK

Input-Process-Output (IPO) Diagram



NEXIAM

Next-Generation
Examinations and
X(Cross)-Role
Intelligent
Admissions
Management

06

/ 14

SCOPE AND LIMITATION

What NEXIAM Covers — And What It Doesn't

SCOPE

- Admission testing workflow only — June to September 2026.
- Seven user roles, full web-based application intake.
- ≥95% target OMR scoring accuracy.
- RAG chatbot, K-Means aptitude dashboard (extensible ML model support), and AI-assisted scheduling — all active.
- The proctoring interface- cloud-connected but offline-resilience caching for intermittent connectivity.

LIMITATION

- No enrollment, tuition, or long-term records.
- Cloud-hosted — proctoring interface alone keeps offline resilience via service-worker caching.
- Free-tier OpenRouter inference caps request volume.
- September evaluation uses proxy evaluators, not live applicants.
- No formal NPC certification — protection is application-layer, not regulatory-certified.

SCOPE

BENEFICIARIES

Who NEXIAM Serves

Institution & Offices

ISPSC, Registrar, Guidance, and Program Heads gain a faster, more compliant, real-time admission process.

Applicants & Students

A 24/7 portal — no repeat trips to campus just to check a result.

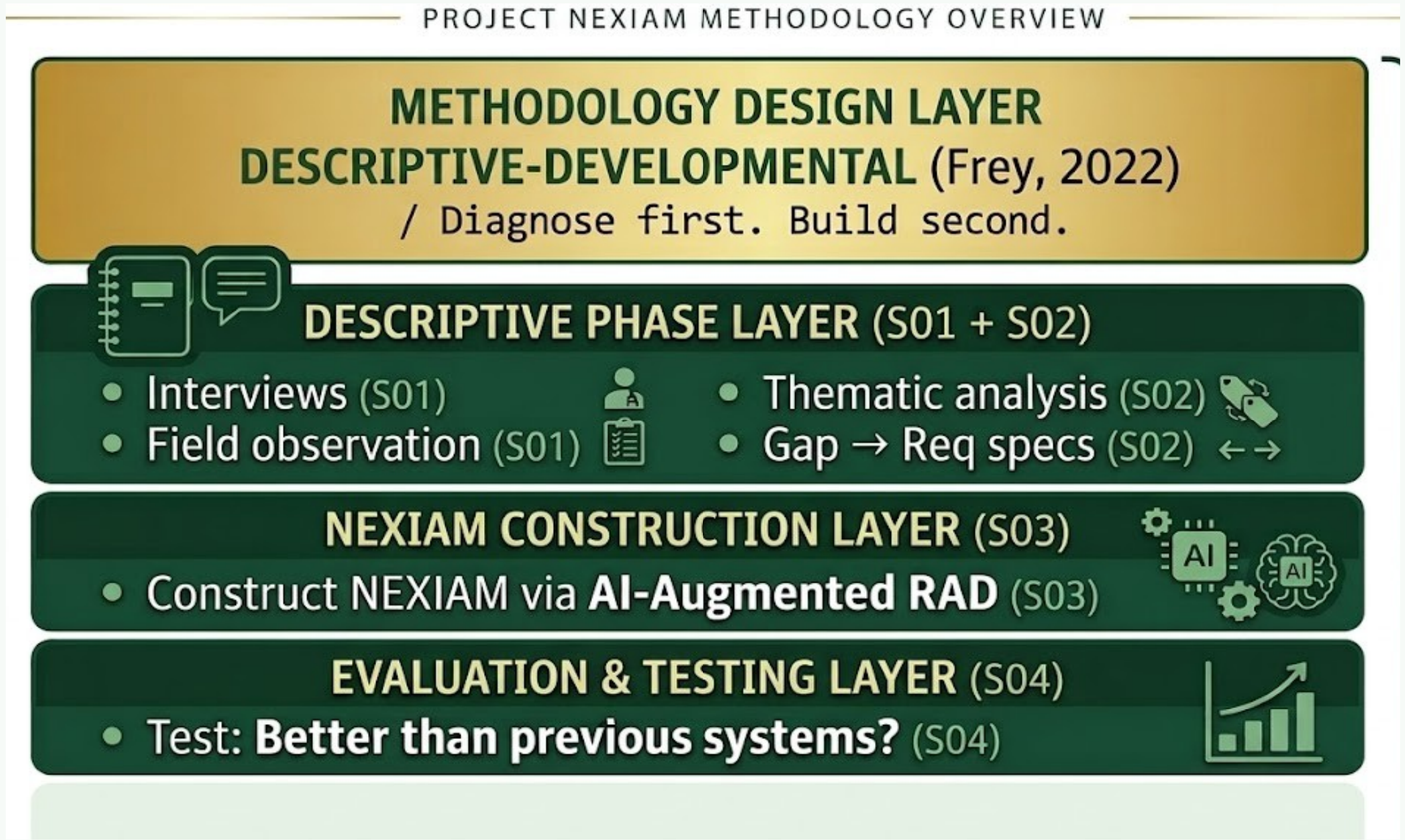
Research Community

Hands-on AI development experience and a blueprint other campuses can replicate.

RESEARCH DESIGN

Descriptive-Developmental, Built Through AI-Augmented RAD

Diagnose the existing process first, then build and evaluate the solution against it.



SOFTWARE MODEL

AI-Augmented Rapid Application Development (RAD)

1 **Requirements Planning**

Map the existing process, gather system requirements.

2 **User Design**

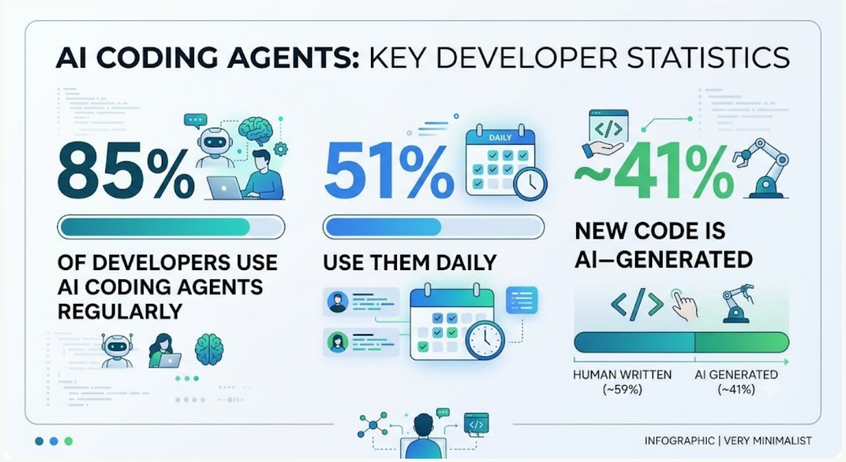
Structure interfaces and workflows around each role.

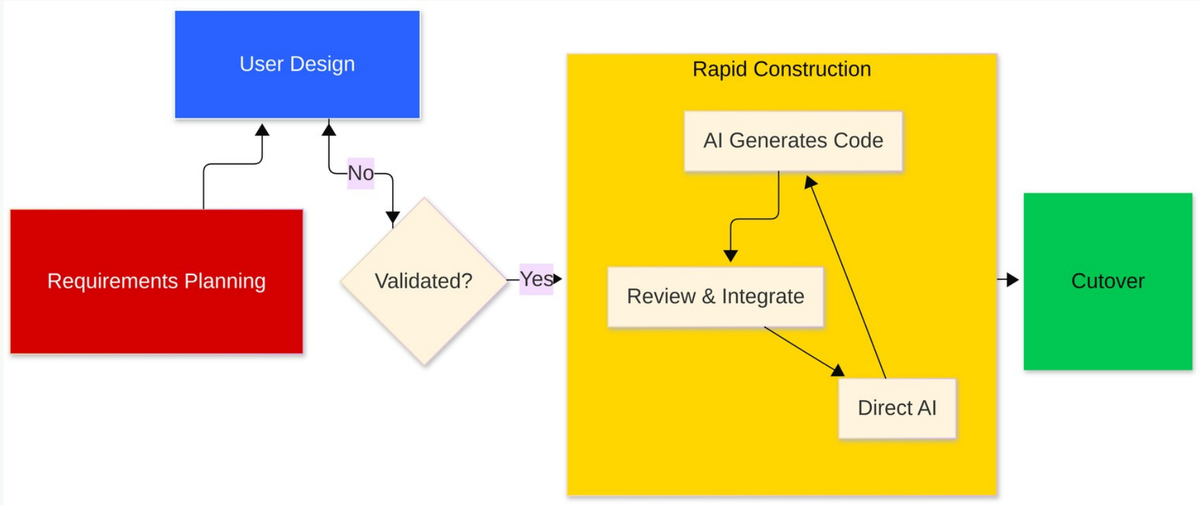
3 **Rapid Construction**

Build the OMR engine, chatbot, and remaining modules.

4 **Cutover**

Deploy, run TAM evaluation, finalize documentation.



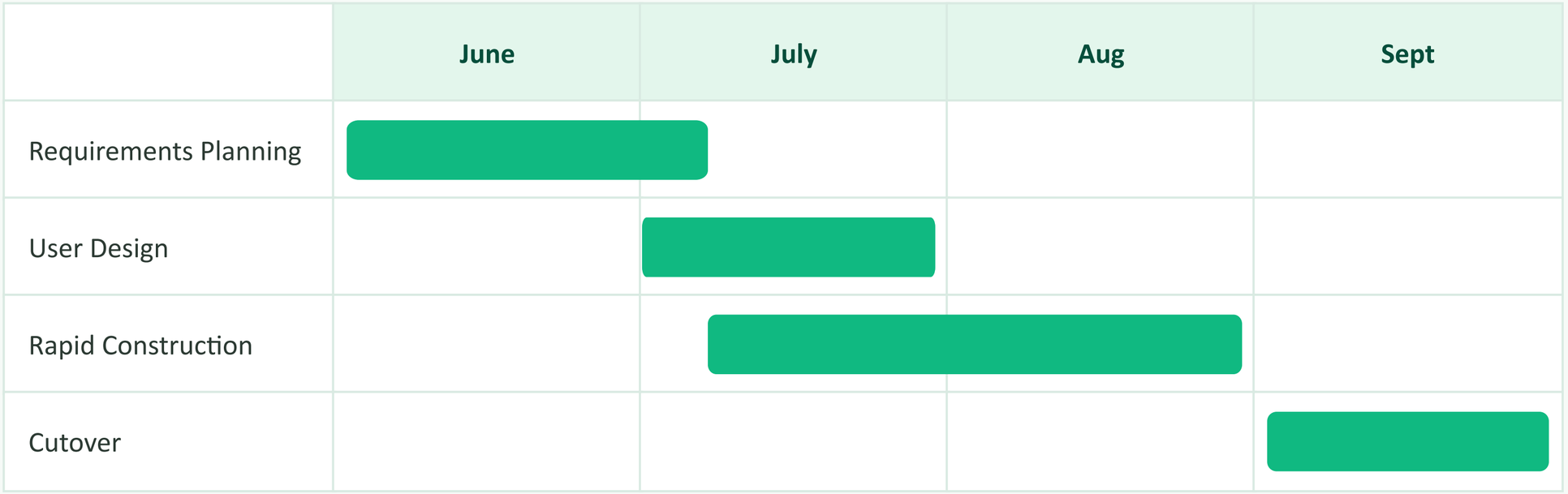


NEXIAM

Next-Generation Examinations and X(Cross)-Role Intelligent Admissions Management

PROJECT PLAN

June – September 2026



On track against this schedule as of today.

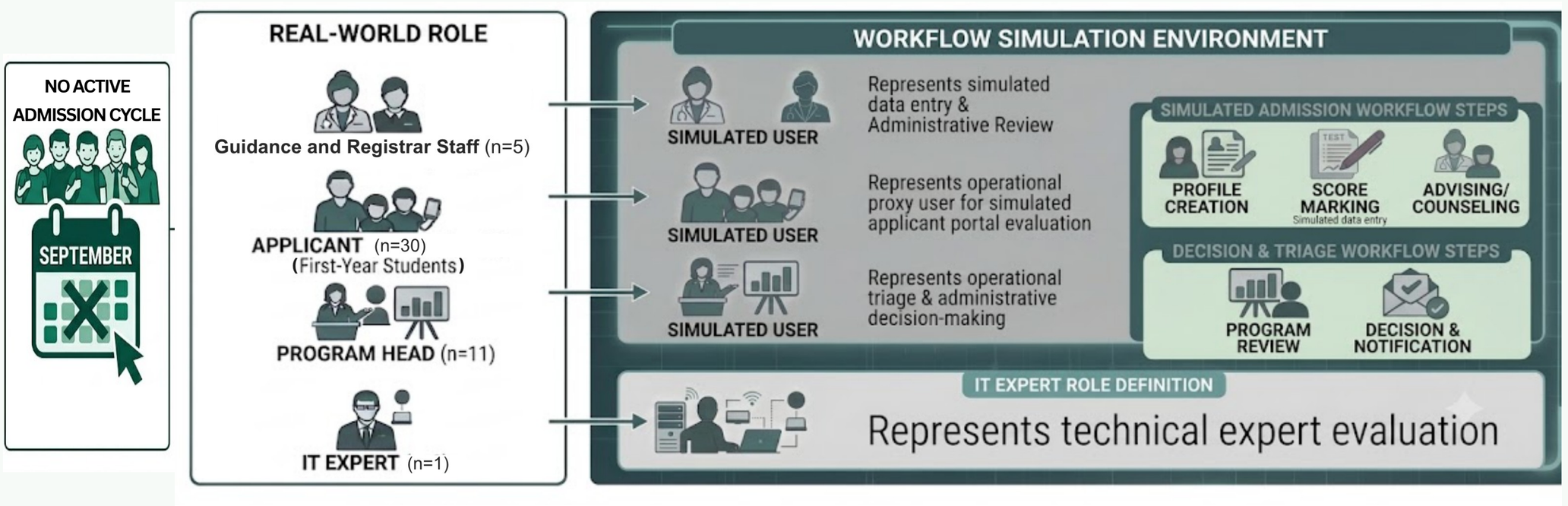
NEXIAM

Next-Generation Examinations and X(Cross)-Role Intelligent Admissions Management

POPULATION AND LOCALE

47 Purposive Evaluators

30	11	3	2	1	47
First-Year Students	Program Heads	Registrar Staff	Guidance Staff	IT Expert	Total



SAMPLE

RESEARCH INSTRUMENT

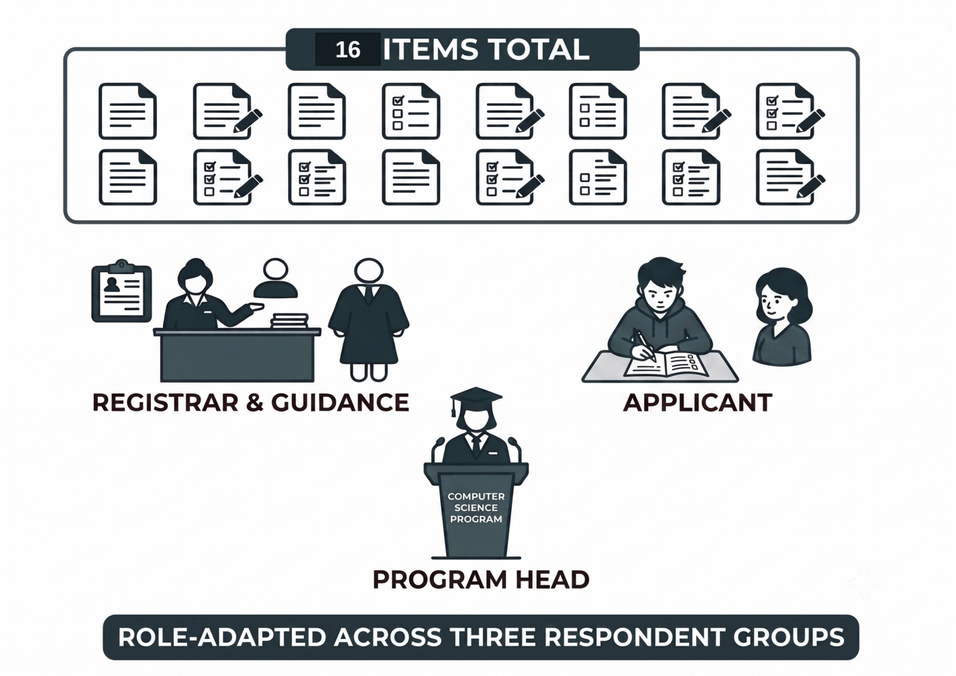
Technology Acceptance Model (Davis, 1989)

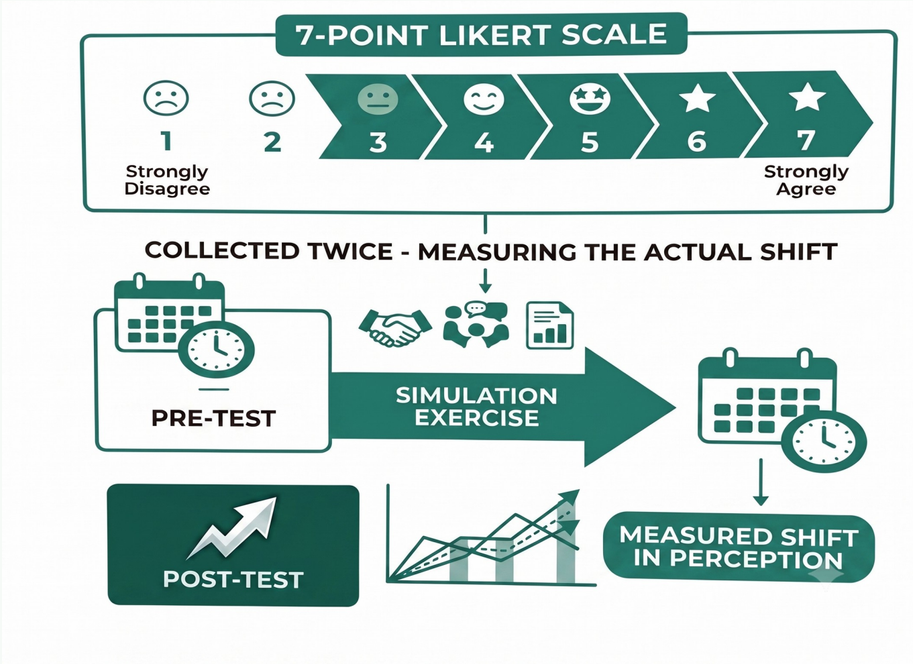
PU
Perceived Usefulness

PEOU
Perceived Ease of Use

ATU
Attitude Toward Use

BIU
Behavioral Intention to Use





DATA ANALYSIS TECHNIQUES

Mapped Directly to the Four Specific Objectives

SO1 — Thematic Analysis

Surfaces the actual pain-point themes from interview and survey responses.

SO2 — Iterative Validation

Proposed requirements confirmed with Registrar and Guidance staff against real workflow.

SO3 — TAM Means

Pooled and per-role construct means, checked against acceptance benchmarks.

SO4 — Shapiro-Wilk → Test

Normality check first, then paired t-test or Wilcoxon at α=.05, with effect size.

NEXIAM · PROPOSAL DEFENSE

Thank You

We're ready for your questions.

SARMIENTO, D. D. · PAGADUAN, J. G. · LOPEZ, C. M.